

PROJECT APEX

Enterprise LNG Operations Intelligence Platform

VOLUME 3

DAX Development

A Complete Measure Library, KPI Definitions, and Advanced Time Intelligence

PART OF THE SERIES

Atlantic LNG Operations Limited (ALOL) — Power BI Project

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Series Overview

Volume 1 — Project Planning & Business Understanding

Volume 2 — Data Preparation & Data Modeling

Volume 3 — DAX Development (this document)

Volume 4 — Dashboard Design & Storytelling

Volume 5 — Portfolio, Documentation & Interview Preparation

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1. Introduction & DAX Strategy

Volume 2 left the ALOL data model fully cleaned, validated, and related — 14 tables, 19 relationships, zero orphaned keys. Volume 3 builds on top of that model the layer that actually answers business questions: the DAX measure library. This volume is organized by subject area (Production, Export, Maintenance, Shutdown, HSE, Finance, Weather) and closes with a cross-functional Advanced DAX section and a single master KPI Definitions table that Volume 4's dashboards will be built directly against.

1.1 Measure Organization Strategy

Every measure in this volume should be created inside a dedicated Measure Table in Power BI, rather than scattered across whichever fact table feels nearest. Create seven empty measure tables (one per subject area below) using Modeling → New Table with a minimal DAX formula such as Production Measures = {BLANK()}, then hide the resulting BLANK column from Report View. This is standard professional practice — it keeps the Fields pane organized and signals clearly to anyone opening the file where to find each calculation.

Measure Table	Houses Measures For
_Production Measures	fact_production
_Export Measures	fact_export_shipments
_Maintenance Measures	fact_maintenance
_Shutdown Measures	fact_shutdown_events
_HSE Measures	fact_hse_incidents
_Finance Measures	fact_finance
_Weather Measures	fact_weather

The leading underscore is a deliberate naming convention — it sorts these measure tables to the top of the Fields pane, visually separating them from the 14 data tables beneath. This is a small detail, but it is exactly the kind of polish that signals professional Power BI practice in a portfolio review or interview.

1.2 DAX Style Conventions Used in This Volume

- All measure names use Title Case with spaces, e.g. Total LNG Production, not TotalLNGProduction
- All column references use fully qualified Table[Column] syntax, never bare column names, even where DAX would allow it — this avoids ambiguity as the model grows
- Every measure includes a one-line comment above the formula in the actual Power BI measure (using // syntax) summarizing its purpose — shown in this document as the description text beneath each formula block

- Variables (VAR) are used for any formula with more than one nested calculation, both for readability and for performance (variables are evaluated once, calculated columns/measures referenced repeatedly are not)

2. Time Intelligence Foundation

Before building subject-area measures, a small set of foundational time intelligence measures should exist that every other measure can build on. These rely on `dim_calendar` being correctly marked as a Date Table, as completed in Volume 2, Section 3.2.

Selected Year

Table: `_Production Measures` (utility measure, usable model-wide)

```
Selected Year = SELECTEDVALUE ( dim_calendar[Year] )
```

Returns the single year currently in filter context, or `BLANK()` if multiple years are selected. Used inside titles and tooltips across Volume 4's dashboards to dynamically label visuals (e.g. 'LNG Production — 2023').

Is Current Year Selected

Table: `_Production Measures` (utility measure)

```
Is Current Year Selected = VAR _SelectedYear = SELECTEDVALUE ( dim_calendar[Year] ) VAR  
_CurrentYear = YEAR ( TODAY ( ) ) RETURN _SelectedYear = _CurrentYear
```

A boolean utility measure used to conditionally format visuals — for example, highlighting the current year's KPI card differently from historical years in the Executive Dashboard.

2.1 Why `dim_calendar` Must Remain the Only Date Table

Power BI's built-in time intelligence functions (`TOTALYTD`, `SAMEPERIODLASTYEAR`, `DATEADD`, and similar) depend entirely on a continuous, gap-free date column marked explicitly as a Date Table. Volume 2 confirmed `dim_calendar` spans every day from 1 January 2019 to 31 December 2024 with no gaps and is marked accordingly. Every time intelligence measure in this volume assumes this foundation; if it is ever broken (for example, by accidentally filtering `dim_calendar` to exclude weekends), every YTD and year-over-year measure in the model will silently return incorrect results.

3. Production Measures

These measures answer the core operational question: how much did we produce, how reliably, and how efficiently. They sit in the _Production Measures table and reference fact_production exclusively unless otherwise noted.

Total LNG Produced

Table: _Production Measures

```
Total LNG Produced = SUM ( fact_production[LNGProduced] )
```

The base aggregation measure for LNG output, in metric tonnes. This is the single most important measure in the entire model and will appear on nearly every dashboard page.

Total LPG Produced

Table: _Production Measures

```
Total LPG Produced = SUM ( fact_production[LPGProduced] )
```

Base aggregation for LPG output (propane and butane combined), in metric tonnes.

Total Condensate Produced

Table: _Production Measures

```
Total Condensate Produced = SUM ( fact_production[CondensateProduced] )
```

Base aggregation for condensate output, in metric tonnes.

Total Gas Received

Table: _Production Measures

```
Total Gas Received = SUM ( fact_production[GasReceived] )
```

Total feed gas intake across all trains, in MMSCF. Used alongside Total LNG Produced to calculate conversion efficiency.

Average Availability %

Table: _Production Measures

```
Average Availability % = AVERAGE ( fact_production[Availability] )
```

The headline reliability KPI for the entire plant. Averaged across all train-days in the current filter context. This measure alone answers ‘how reliable was the plant this year/this train/this month’ and is the primary KPI card on the Executive and Production dashboards.

Average Utilization %

Table: _Production Measures

```
Average Utilization % = AVERAGE ( fact_production[Utilization] )
```

Companion measure to Availability — Utilization captures how much of the available capacity was actually used, distinguishing ‘the train was online’ from ‘the train was running at full output while online’.

Conversion Efficiency %

Table: _Production Measures

```
Conversion Efficiency % = VAR _LNG = [Total LNG Produced] VAR _Gas = [Total Gas Received]
RETURN DIVIDE ( _LNG, _Gas, BLANK() )
```

The ratio of LNG output to feed gas input — a core process-engineering efficiency metric. Uses DIVIDE() rather than the / operator throughout this entire volume specifically to gracefully return BLANK() instead of an error when the denominator is zero (for example, when a train is fully shut down and Gas Received is 0 for that period).

Total Downtime Hours

Table: _Production Measures

```
Total Downtime Hours = VAR _AvailHours = SUMX ( fact_production, ( 1 -
fact_production[Availability] / 100 ) * 24 ) RETURN _AvailHours
```

Estimates total hours of downtime implied by availability shortfall, assuming a 24-hour operating day. This is an estimation measure, not a measure of recorded outage events (those are counted directly in Section 6, Shutdown Measures) — the distinction should be explained explicitly on any dashboard tooltip that surfaces this measure, since stakeholders may otherwise assume it reconciles exactly with fact_shutdown_events.

Total Operating Cost

Table: _Production Measures

```
Total Operating Cost = SUM ( fact_production[OperatingCost] )
```

Daily operating cost summed across the current filter context, in USD.

LNG Produced YTD

Table: _Production Measures

```
LNG Produced YTD = TOTALYTD ( [Total LNG Produced], dim_calendar[Date] )
```

Year-to-date cumulative LNG production, using native Power BI time intelligence against the marked Date Table. Resets automatically at the start of each calendar year as the filter context moves.

LNG Produced PY

Table: _Production Measures

```
LNG Produced PY = CALCULATE ( [Total LNG Produced], SAMEPERIODLASTYEAR (
dim_calendar[Date] ) )
```

Prior-year LNG production for the same period currently in filter context — the base comparison measure that the YoY% measure below is built from.

LNG Production YoY %

Table: _Production Measures

```
LNG Production YoY % = VAR _Current = [Total LNG Produced] VAR _Prior = [LNG Produced PY]
RETURN DIVIDE ( _Current - _Prior, _Prior, BLANK() )
```

Year-over-year percentage growth in LNG production. This is the headline trend KPI on the Executive Dashboard, typically displayed with a conditional up/down arrow.

Production Rank by Train

Table: _Production Measures

```
Production Rank by Train = RANKX ( ALL ( dim_trains[TrainName] ), [Total LNG
Produced], , DESC )
```

Ranks trains 1 through 7 by total LNG output within the current filter context (e.g. within the selected year). Used to power a dynamic ‘Top Performing Train’ visual on the Production Dashboard.

4. Export Measures

These measures answer commercial questions: how much did we sell, to whom, at what price, and how profitably. They reference fact_export_shipments and sit in the _Export Measures table.

Total Shipments

Table: _Export Measures

```
Total Shipments = CALCULATE ( COUNTROWS ( fact_export_shipments ),
fact_export_shipments[ShipmentStatus] <> "Cancelled" )
```

Counts shipments excluding Cancelled status, consistent with the business rule established in Volume 2, Section 3.10. This is the default shipment-count measure used across all dashboards; a separate Cancelled Shipments measure (below) tracks cancellations explicitly rather than silently dropping them from the model.

Cancelled Shipments

Table: _Export Measures

```
Cancelled Shipments = CALCULATE ( COUNTROWS ( fact_export_shipments ),
fact_export_shipments[ShipmentStatus] = "Cancelled" )
```

Explicit count of cancelled shipments, surfaced separately so cancellations remain visible and trackable rather than disappearing from reporting entirely.

Total Export Revenue

Table: _Export Measures

```
Total Export Revenue = CALCULATE ( SUM ( fact_export_shipments[Revenue] ),
fact_export_shipments[ShipmentStatus] <> "Cancelled" )
```

Total realized revenue from delivered, in-transit, and loading shipments, in USD. Excludes Cancelled shipments since no revenue is realized on a cancelled cargo.

Total Freight Cost

Table: _Export Measures

```
Total Freight Cost = SUM ( fact_export_shipments[FreightCost] )
```

Total cost of shipping across all cargoes, in USD.

Net Revenue After Freight

Table: _Export Measures

```
Net Revenue After Freight = [Total Export Revenue] - [Total Freight Cost]
```

Revenue net of shipping cost — a more realistic commercial profitability measure than gross revenue alone.

Average Price per Unit

Table: _Export Measures

```
Average Price per Unit = AVERAGEX ( fact_export_shipments,
fact_export_shipments[UnitPrice] )
```

Simple average unit price across shipments in the current filter context. Useful for spotting pricing trends by product or by customer over time.

Total Cargo Quantity

Table: _Export Measures

```
Total Cargo Quantity = SUM ( fact_export_shipments[Quantity] )
```

Total volume shipped, in the relevant product unit. Note this measure should not be summed blindly across multiple products at once, since LNG, LPG, and Condensate use different units — always pair this measure with a ProductName filter or slicer on dashboards (addressed again in Volume 4).

Distinct Customers Served

Table: _Export Measures

```
Distinct Customers Served = DISTINCTCOUNT ( fact_export_shipments[CustomerID] )
```

Counts unique customers with at least one shipment in the current filter context. Used to track customer base breadth over time and by region.

Top Customer by Revenue

Table: _Export Measures

```
Top Customer by Revenue = VAR _TopCustomer = TOPN ( 1, VALUES (
dim_customers[CustomerName] ), [Total Export Revenue], DESC ) RETURN
CONCATENATEX ( _TopCustomer, dim_customers[CustomerName] )
```

Dynamically returns the name of the highest-revenue customer in the current filter context. Used in a KPI card on the Export Dashboard.

Customer Concentration % (Top 5)

Table: _Export Measures

```
Customer Concentration % (Top 5) = VAR _Top5Revenue = SUMX ( TOPN ( 5, VALUES
( dim_customers[CustomerID] ), [Total Export Revenue], DESC ), [Total Export
Revenue] ) VAR _TotalRevenue = [Total Export Revenue] RETURN DIVIDE ( _Top5Revenue,
_TotalRevenue, BLANK() )
```

Measures revenue concentration risk — what percentage of total revenue comes from just the top 5 customers. A high percentage signals commercial risk from over-reliance on a small customer base, a genuinely important risk metric for any export business.

Export Revenue YoY %

Table: _Export Measures

```
Export Revenue YoY % = VAR _Current = [Total Export Revenue] VAR _Prior = CALCULATE (
[Total Export Revenue], SAMEPERIODLASTYEAR ( dim_calendar[Date] ) ) RETURN DIVIDE (
_Current - _Prior, _Prior, BLANK() )
```

Year-over-year revenue growth, following the same pattern established in Section 3 for production.

5. Maintenance Measures

These measures support the Maintenance Manager persona defined in Volume 1, Section 6 — tracking work order volume, cost, and downtime by equipment, train, and maintenance type.

Total Work Orders

Table: _Maintenance Measures

```
Total Work Orders = COUNTROWS ( fact_maintenance )
```

Total count of maintenance work orders in the current filter context, regardless of status.

Completed Work Orders

Table: _Maintenance Measures

```
Completed Work Orders = CALCULATE ( COUNTROWS ( fact_maintenance ),  
fact_maintenance[Status] = "Completed" )
```

Count of work orders with Completed status — used to calculate completion rate below.

Work Order Completion Rate %

Table: _Maintenance Measures

```
Work Order Completion Rate % = DIVIDE ( [Completed Work Orders], [Total Work Orders],  
BLANK() )
```

Percentage of maintenance work orders successfully completed, a basic operational health indicator for the Maintenance department.

Total Maintenance Cost

Table: _Maintenance Measures

```
Total Maintenance Cost = SUM ( fact_maintenance[MaintenanceCost] )
```

Total spend on maintenance activity, in USD, across the current filter context.

Total Maintenance Downtime Hours

Table: _Maintenance Measures

```
Total Maintenance Downtime Hours = SUM ( fact_maintenance[DowntimeHours] )
```

Sum of downtime hours directly attributed to maintenance work orders — distinct from the estimated downtime in Section 3, since this figure is tied to actual logged work orders rather than an availability-based estimate.

Average Cost per Work Order

Table: _Maintenance Measures

```
Average Cost per Work Order = DIVIDE ( [Total Maintenance Cost], [Total Work Orders],  
BLANK() )
```

Average cost per maintenance activity — useful for spotting cost trends by maintenance type or by equipment category.

Corrective Maintenance Cost

Table: _Maintenance Measures

```
Corrective Maintenance Cost = CALCULATE ( [Total Maintenance Cost],
fact_maintenance[MaintenanceType] = "Corrective" )
```

Isolates spend specifically on corrective (reactive/breakdown) maintenance, as opposed to preventive or predictive work — a key reliability indicator, since a rising share of corrective spend signals a deteriorating, reactive maintenance posture.

Corrective Maintenance Share %

Table: _Maintenance Measures

```
Corrective Maintenance Share % = DIVIDE ( [Corrective Maintenance Cost], [Total
Maintenance Cost], BLANK() )
```

What percentage of total maintenance spend is reactive rather than planned. This is one of the most commonly tracked reliability KPIs in real industrial maintenance functions — a healthy plant typically targets a low corrective share, with the majority of spend going to preventive and predictive work instead.

High Criticality Equipment Maintenance Cost

Table: _Maintenance Measures

```
High Criticality Equipment Maintenance Cost = CALCULATE ( [Total Maintenance Cost],
dim_equipment[Criticality] IN { "Critical A", "Critical B" } )
```

Isolates spend on the plant's most critical equipment categories, allowing the Maintenance Manager to confirm that the highest-risk assets are receiving proportionate attention. This measure relies on the relationship from dim_equipment to fact_maintenance built in Volume 2.

6. Shutdown Measures

These measures are the most technically involved in the entire library, because `fact_shutdown_events` carries the dual-date relationship established in Volume 2, Section 5.3. Read that section again before working through this one if it is not fresh in memory — every `USERRELATIONSHIP()` measure below depends on understanding which relationship is active by default.

6.1 Recap: Active vs. Inactive Relationship

`dim_calendar[DateKey] → fact_shutdown_events[StartDateKey]` is the active relationship.

`dim_calendar[DateKey] → fact_shutdown_events[EndDateKey]` exists but is inactive. This means that, by default, any visual filtered by date will filter shutdown events by when they started. Measures below that need to filter by when shutdowns ended must explicitly activate the inactive relationship using `USERRELATIONSHIP()`.

Total Shutdown Events

Table: `_Shutdown Measures`

```
Total Shutdown Events = COUNTROWS ( fact_shutdown_events )
```

Total count of shutdown events in the current filter context. Uses the active (Start Date) relationship by default — this counts shutdowns that started within the selected period.

Shutdown Events Ended in Period

Table: `_Shutdown Measures`

```
Shutdown Events Ended in Period = CALCULATE (
    COUNTROWS ( fact_shutdown_events ),
    USERRELATIONSHIP ( dim_calendar[DateKey], fact_shutdown_events[EndDateKey] )
)
```

Counts shutdowns that concluded within the selected period, explicitly activating the otherwise-inactive `EndDateKey` relationship. This is the canonical, textbook use case for `USERRELATIONSHIP()` and is worth being able to explain confidently — Volume 5's interview preparation guide returns to this exact measure as a worked example.

Total Lost Production (Shutdowns)

Table: `_Shutdown Measures`

```
Total Lost Production (Shutdowns) = SUM ( fact_shutdown_events[LostProduction] )
```

Sum of estimated production lost to shutdown events, in metric tonnes, using the active (start date) relationship by default.

Total Shutdown Cost

Table: `_Shutdown Measures`

```
Total Shutdown Cost = SUM ( fact_shutdown_events[ShutdownCost] )
```

Total cost associated with shutdown events, in USD, including both direct cost and estimated lost revenue as captured at the data engineering stage.

Average Shutdown Duration (Days)

Table: _Shutdown Measures

```
Average Shutdown Duration (Days) = AVERAGE ( fact_shutdown_events[DurationDays] )
```

Average shutdown length across the current filter context — typically sliced by ShutdownType, since Planned Turnarounds (21–35 days) and Emergency Shutdowns (1–10 days) have very different natural durations and should rarely be averaged together without that context.

Planned Turnaround Count

Table: _Shutdown Measures

```
Planned Turnaround Count = CALCULATE ( [Total Shutdown Events],  
fact_shutdown_events[ShutdownType] = "Planned Turnaround" )
```

Isolates planned turnaround events specifically. In the current dataset, 120 of the 227 total shutdown events are Planned Turnarounds — slightly over half — which is a reasonable, realistic balance to highlight when presenting this measure.

Emergency Shutdown Count

Table: _Shutdown Measures

```
Emergency Shutdown Count = CALCULATE ( [Total Shutdown Events],  
fact_shutdown_events[ShutdownType] = "Emergency Shutdown" )
```

Isolates unplanned emergency shutdown events. A rising trend in this measure, particularly relative to Planned Turnaround Count, is a leading indicator of declining plant reliability worth surfacing prominently on the Shutdown Dashboard.

Unplanned Shutdown Share %

Table: _Shutdown Measures

```
Unplanned Shutdown Share % = VAR _Unplanned = CALCULATE ( [Total Shutdown  
Events], fact_shutdown_events[ShutdownType] IN { "Emergency Shutdown", "Feed Gas  
Interruption", "Partial Curtailment" } ) VAR _Total = [Total Shutdown Events] RETURN  
DIVIDE ( _Unplanned, _Total, BLANK() )
```

What percentage of all shutdown events were unplanned (anything other than a scheduled Planned Turnaround). This is the single most important reliability KPI for the Shutdown Dashboard's executive summary card.

7. HSE Measures

These measures support the HSE Manager persona and the compliance reporting need identified in Volume 1. They reference fact_hse_incidents.

Total HSE Incidents

Table: _HSE Measures

```
Total HSE Incidents = COUNTROWS ( fact_hse_incidents )
```

Total incident count in the current filter context, across all severities and types.

Critical Incidents

Table: _HSE Measures

```
Critical Incidents = CALCULATE ( [Total HSE Incidents],  
fact_hse_incidents[Severity] = "Critical" )
```

Isolates only Critical-severity incidents — the smallest but highest-impact category, deserving prominent, dedicated visibility on the HSE Dashboard rather than being buried inside an aggregate count.

Total Lost Time Hours (HSE)

Table: _HSE Measures

```
Total Lost Time Hours (HSE) = SUM ( fact_hse_incidents[LostTimeHours] )
```

Sum of work hours lost to HSE incidents — a standard safety KPI tracked by essentially every industrial operator. Named explicitly with '(HSE)' suffix to avoid any naming collision with downtime measures from Sections 3 and 5.

Lost Time Injury Frequency Rate

Table: _HSE Measures

```
Lost Time Injury Frequency Rate = VAR _LTI = CALCULATE ( [Total HSE  
Incidents], fact_hse_incidents[IncidentType] = "Lost Time Injury" ) VAR  
_HoursWorked = 1000000 RETURN DIVIDE ( _LTI * 200000, _HoursWorked, BLANK() )
```

Models the industry-standard LTIFR calculation (incidents per 200,000 hours worked, the OSHA-style convention used across the global oil & gas sector). Note this measure uses a placeholder constant for total hours worked since fact_employees does not currently track actual hours worked per period — this limitation should be stated explicitly on any dashboard using this measure, and flagged as a genuine data enhancement opportunity in Volume 5's documentation rather than presented as a fully precise figure.

Incidents per Equipment Item

Table: _HSE Measures

```
Incidents per Equipment Item = DIVIDE ( [Total HSE Incidents], DISTINCTCOUNT ( fact_hse_incidents[EquipmentID] ), BLANK() )
```


Average incident count per unique piece of equipment involved — useful for identifying whether incidents are concentrated on a small number of problem assets or spread broadly across the plant.

HSE Incident YoY %

Table: _HSE Measures

```
HSE Incident YoY % = VAR _Current = [Total HSE Incidents] VAR _Prior = CALCULATE ( [Total HSE Incidents], SAMEPERIODLASTYEAR ( dim_calendar[Date] ) ) RETURN DIVIDE ( _Current - _Prior, _Prior, BLANK() )
```

Year-over-year change in incident count. Unlike the Revenue and Production YoY measures, a negative value here is the desired direction — this should be called out explicitly in any Volume 4 visual that uses conditional formatting, since the standard ‘green up / red down’ convention used elsewhere in the model must be deliberately reversed for this specific measure.

8. Finance Measures

These measures serve the CFO persona and directly support the Volume 1 objective of linking operational performance to financial outcomes. They reference fact_finance.

Total Revenue

Table: _Finance Measures

```
Total Revenue = SUM ( fact_finance[Revenue] )
```

Total recognized revenue across the current filter context, in USD. Note this is distinct from Total Export Revenue (Section 4), which is derived from individual shipment transactions — fact_finance represents the consolidated general-ledger view, while fact_export_shipments represents the operational/commercial transaction view. Both should reconcile approximately but are intentionally separate fact tables, consistent with how a real ERP and a real trading/shipping system would differ.

Total OPEX

Table: _Finance Measures

```
Total OPEX = SUM ( fact_finance[OPEX] )
```

Total operating expenditure, in USD.

Total CAPEX

Table: _Finance Measures

```
Total CAPEX = SUM ( fact_finance[CAPEX] )
```

Total capital expenditure, in USD.

Total Profit

Table: _Finance Measures

```
Total Profit = SUM ( fact_finance[Profit] )
```

Total profit, in USD, already pre-calculated at the data engineering stage as Revenue – OPEX – CAPEX – MaintenanceCost (confirmed in Volume 2, Section 3.14). This measure simply aggregates the stored value rather than recalculating it, which is the correct approach since the underlying calculation was already validated row-by-row during data preparation.

Profit Margin %

Table: _Finance Measures

```
Profit Margin % = DIVIDE ( [Total Profit], [Total Revenue], BLANK() )
```

Profit as a percentage of revenue — the headline profitability KPI on the Executive and Finance dashboards.

OPEX as % of Revenue

Table: _Finance Measures

```
OPEX as % of Revenue = DIVIDE ( [Total OPEX], [Total Revenue], BLANK() )
```

Cost-efficiency ratio tracking operating expenditure relative to revenue scale.

Profit YTD

Table: _Finance Measures

```
Profit YTD = TOTALYTD ( [Total Profit], dim_calendar[Date] )
```

Year-to-date cumulative profit, following the same native time intelligence pattern used in Section 3.

Profit YoY %

Table: _Finance Measures

```
Profit YoY % = VAR _Current = [Total Profit] VAR _Prior = CALCULATE ( [Total Profit],  
SAMEPERIODLASTYEAR ( dim_calendar[Date] ) ) RETURN DIVIDE ( _Current - _Prior, _Prior,  
BLANK() )
```

Year-over-year profit growth, the single most-watched figure on the Executive Dashboard.

Revenue per Tonne LNG

Table: _Finance Measures

```
Revenue per Tonne LNG = DIVIDE ( [Total Revenue], [Total LNG Produced], BLANK() )
```

A genuine cross-subject-area measure — combining fact_finance with a measure built on fact_production via Section 3. This works correctly in the model specifically because both fact tables relate to the same shared dim_calendar and dim_products dimensions, exactly the cross-functional analytical capability the star schema design in Volume 1 and Volume 2 was built to enable.

9. Weather Measures

These measures are smaller in number but support the Logistics/Shipping Coordinator persona and the weather-to-operations linkage identified as Business Problem 6 in Volume 1. They reference `fact_weather`.

Average Temperature

Table: `_Weather Measures`

```
Average Temperature = AVERAGE ( fact_weather[Temperature] )
```

Average daily temperature in degrees Celsius across the current filter context.

Total Rainfall

Table: `_Weather Measures`

```
Total Rainfall = SUM ( fact_weather[Rainfall] )
```

Cumulative rainfall in millimetres across the current filter context — useful for identifying the rainy season's operational impact when cross-filtered against production or shipment measures.

Average Wind Speed

Table: `_Weather Measures`

```
Average Wind Speed = AVERAGE ( fact_weather[WindSpeed] )
```

Average daily wind speed in km/h — relevant to marine loading safety windows at the jetty.

High Wind Day Count

Table: `_Weather Measures`

```
High Wind Day Count = CALCULATE ( DISTINCTCOUNT ( fact_weather[DateKey] ),  
fact_weather[WindSpeed] > 30 )
```

Counts days where wind speed exceeded 30 km/h — a threshold commonly used in real marine loading operations to flag potentially unsafe jetty conditions. The specific threshold value should be confirmed against ALOL's (fictional) marine operations procedure when this measure is implemented, and is flagged here as configurable rather than fixed.

10. Advanced DAX Measures

This section collects measures that are either genuinely cross-functional (spanning multiple fact tables), use more advanced DAX patterns (parent-child style ranking, dynamic titles, what-if parameters), or exist specifically to demonstrate technical depth in a portfolio context.

Estimated Revenue Lost to Shutdowns

Table: _Shutdown Measures

```
Estimated Revenue Lost to Shutdowns = VAR _LostTonnes = [Total Lost Production
(Shutdowns)] VAR _AvgPricePerTonne = DIVIDE ( CALCULATE ( [Total Export
Revenue], ALL ( dim_calendar ) ), CALCULATE ( [Total Cargo Quantity], ALL (
dim_calendar ) ), BLANK() ) RETURN _LostTonnes * _AvgPricePerTonne
```

Directly answers the question posed in Volume 1, Section 4, Problem 2: ‘how much revenue did we lose due to a shutdown.’ This measure deliberately spans three fact tables conceptually — shutdown events, production loss, and export pricing — to estimate financial impact. The ALL(dim_calendar) inside the average price calculation intentionally uses the all-time average price per tonne, rather than the currently filtered period's price, since using a filtered price would create a circular, self-referential estimate when this measure is itself sliced by date. This nuance is worth being able to explain in an interview — it demonstrates active reasoning about filter context, not just formula syntax.

Dynamic KPI Title — Selected Train

Table: _Production Measures

```
Dynamic KPI Title – Selected Train = VAR _TrainCount = DISTINCTCOUNT (
dim_trains[TrainName] ) VAR _SelectedTrain = SELECTEDVALUE ( dim_trains[TrainName] )
RETURN IF ( _TrainCount = 1, "Production Performance – " & _SelectedTrain,
"Production Performance – All Trains" )
```

A dynamic title measure bound directly to a Card or Text visual's title field, so the Production Dashboard's header text changes automatically as a user slices by train. This pattern is used extensively in Volume 4's wireframes to make every dashboard page feel responsive and intentional rather than static.

Equipment Reliability Score

Table: _Maintenance Measures

```
Equipment Reliability Score = VAR _IncidentCount = CALCULATE ( [Total HSE Incidents],
ALLEXCEPT ( dim_equipment, dim_equipment[EquipmentID] ) ) VAR _MaintenanceCount =
CALCULATE ( [Total Work Orders], ALLEXCEPT ( dim_equipment, dim_equipment[EquipmentID] ) )
VAR _CorrectiveCount = CALCULATE ( [Total Work Orders],
fact_maintenance[MaintenanceType] = "Corrective", ALLEXCEPT ( dim_equipment,
dim_equipment[EquipmentID] ) ) RETURN 100 - ( ( _IncidentCount * 5 ) + (
_CorrectiveCount * 2 ) )
```

A composite, weighted scoring measure combining incident history and corrective maintenance frequency into a single 0–100 reliability score per equipment item, using ALLEXCEPT to force the calculation to evaluate per-equipment regardless of any other filters (date, train) currently applied elsewhere on the page. The specific weights (5 points per incident, 2 points per corrective work order)

are illustrative starting values and should be calibrated against real stakeholder input if this were a live engagement — this is noted explicitly here rather than presented as a definitive industry formula, since composite scoring weights are always a business judgment call, not a technical one.

Shutdown Cost as % of Total Finance Cost Base

Table: _Shutdown Measures

```
Shutdown Cost as % of Total Finance Cost Base = VAR _ShutdownCost = [Total Shutdown Cost]
VAR _TotalCostBase = CALCULATE ( [Total OPEX] + [Total CAPEX] + [Total
Maintenance Cost], ALL ( fact_shutdown_events ) ) RETURN DIVIDE (
_ShutdownCost, _TotalCostBase, BLANK() )
```

Places shutdown cost in context against the company's broader cost base, helping the CFO persona understand shutdown impact proportionally rather than as an isolated absolute figure.

11. KPI Definitions Library

This master table consolidates the most important measures from every section above into a single reference, organized by dashboard. Volume 4 will use this table directly as the specification for which KPI cards appear on which dashboard page.

11.1 Executive Dashboard KPIs

KPI	Measure Used	Target Direction
Total LNG Produced (YTD)	LNG Produced YTD	Higher is better
Plant Availability %	Average Availability %	Higher is better
Total Revenue	Total Revenue	Higher is better
Profit Margin %	Profit Margin %	Higher is better
Total HSE Incidents	Total HSE Incidents	Lower is better
Unplanned Shutdown Share %	Unplanned Shutdown Share %	Lower is better

11.2 Production Dashboard KPIs

KPI	Measure Used	Target Direction
Total LNG Produced	Total LNG Produced	Higher is better
Conversion Efficiency %	Conversion Efficiency %	Higher is better
Average Utilization %	Average Utilization %	Higher is better
Production YoY %	LNG Production YoY %	Higher is better
Top Performing Train	Production Rank by Train	N/A — informational

11.3 Export Dashboard KPIs

KPI	Measure Used	Target Direction
Total Export Revenue	Total Export Revenue	Higher is better
Total Shipments	Total Shipments	Higher is better
Customer Concentration % (Top 5)	Customer Concentration % (Top 5)	Lower is better (risk)
Average Price per Unit	Average Price per Unit	Context-dependent
Export Revenue YoY %	Export Revenue YoY %	Higher is better

11.4 Maintenance Dashboard KPIs

KPI	Measure Used	Target Direction
Total Maintenance Cost	Total Maintenance Cost	Lower is better, in proportion
Corrective Maintenance Share %	Corrective Maintenance Share %	Lower is better
Work Order Completion Rate %	Work Order Completion Rate %	Higher is better
Equipment Reliability Score	Equipment Reliability Score	Higher is better

11.5 Shutdown Dashboard KPIs

KPI	Measure Used	Target Direction
Total Shutdown Events	Total Shutdown Events	Lower is better
Unplanned Shutdown Share %	Unplanned Shutdown Share %	Lower is better
Estimated Revenue Lost to Shutdowns	Estimated Revenue Lost to Shutdowns	Lower is better
Average Shutdown Duration (Days)	Average Shutdown Duration (Days)	Lower is better (by type)

11.6 HSE Dashboard KPIs

KPI	Measure Used	Target Direction
Total HSE Incidents	Total HSE Incidents	Lower is better
Critical Incidents	Critical Incidents	Lower is better
Lost Time Injury Frequency Rate	Lost Time Injury Frequency Rate	Lower is better
HSE Incident YoY %	HSE Incident YoY %	Lower is better (reversed convention)

11.7 Finance Dashboard KPIs

KPI	Measure Used	Target Direction
Total Revenue	Total Revenue	Higher is better
Total Profit	Total Profit	Higher is better
Profit Margin %	Profit Margin %	Higher is better
OPEX as % of Revenue	OPEX as % of Revenue	Lower is better
Profit YoY %	Profit YoY %	Higher is better

11.8 Weather Dashboard KPIs

KPI	Measure Used	Target Direction
Average Temperature	Average Temperature	Informational
Total Rainfall	Total Rainfall	Informational
High Wind Day Count	High Wind Day Count	Lower is better (for marine ops)

12. DAX Measure Validation & Testing

Every measure in this volume must be spot-checked before being trusted on a dashboard. This is not optional polish — an unvalidated measure looks identical to a correct one until someone makes a decision based on a wrong number.

12.1 Validation Checklist

- ☐ For every SUM-based measure, manually total a small filtered subset (e.g. one train, one month) in Excel or a simple Power Query reference, and compare against the Power BI card visual
- ☐ For every ratio/percentage measure using DIVIDE(), confirm it returns BLANK() rather than an error or infinity when the denominator is filtered to zero (test this explicitly by filtering to a period with no data)
- ☐ For both USERELATIONSHIP() measures in Section 6, build a simple test table visual with DateKey, Total Shutdown Events, and Shutdown Events Ended in Period side by side, and manually confirm at least 3 known shutdown records appear in the correct measure based on their actual start and end dates
- ☐ For every YoY% measure, confirm the very first year in the dataset (2019) correctly returns BLANK() rather than a misleading 0% or error, since there is no 2018 data to compare against
- ☐ For Customer Concentration % (Top 5) and similar TOPN-based measures, manually rank the underlying data in Power Query or Excel for at least one period and confirm the same top 5 customers and the same percentage are returned
- ☐ Cross-check Total Revenue (fact_finance) against Total Export Revenue (fact_export_shipments) at the annual level — they should track closely (both represent revenue) even though they are independently sourced fact tables, and any large, unexplained divergence should be investigated and documented rather than ignored

12.2 Common DAX Pitfalls Specifically Relevant to This Model

- Forgetting DIVIDE() and using the / operator directly — causes visible #DIV/0! errors in dashboards rather than clean blanks, and should never appear in any finished measure in this model
- Applying a measure built for fact_production (e.g. Conversion Efficiency %) to a visual sliced only by dim_employees or dim_customers, where the relationship chain does not actually connect — Power BI will not error, but will silently return a meaningless or blank result; always confirm a measure's relevant dimension before placing it on a new visual
- Forgetting that Total Cargo Quantity sums different units across products (MMBTU, MT, BBL) if not filtered by a single product — always pair this measure with a ProductName context as documented in Section 4

- Using the inactive EndDateKey relationship by default without realizing it is inactive, and being confused when a measure intended to filter by end date silently returns the same result as the start-date-filtered version — this is exactly why Section 6.1 exists as a standing recap

What Comes Next: Volume 4 Preview

With a validated, organized DAX measure library now complete — seven measure tables, more than 45 individual measures, and a master KPI library mapped to all nine planned dashboard pages — Volume 4: Dashboard Design & Storytelling will take this measure library and turn it into the actual visual product. Specifically, Volume 4 will cover:

- Page-by-page wireframes for all nine dashboards (Executive, Production, Export, Maintenance, Shutdown, HSE, Finance, Weather, Map)
- The full UI/UX system: color palette, typography, layout grid, and navigation plan
- Drill-through design connecting summary dashboards to detail pages
- Tooltip design standards using several of the measures built in this volume
- A storytelling guide explaining how to present this dashboard suite to the stakeholders defined in Volume 1

Every wireframe and KPI card placement in Volume 4 will reference the exact measure names defined in this volume, so this document should remain open alongside Volume 4 as a lookup reference once that volume is built.

End of Volume 3 — DAX Development